Applied NLP

DATA 641

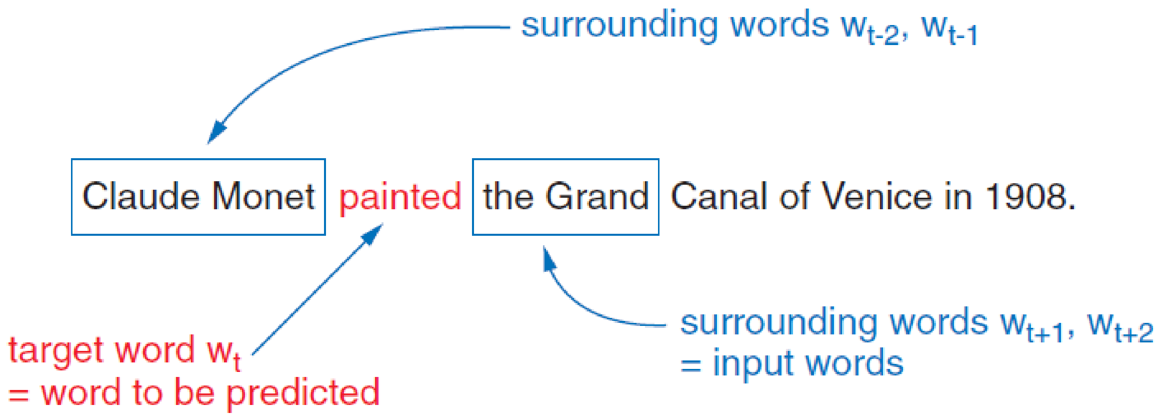
Lecture 5 — Text Representation; CBOW

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, ISBN 9781617294631

Continuous Bag-of-Words approach

- **Continuous bag of words (CBOW)**: Predict the "center" word from the words in its context

In the continuous bag-of-words approach, we are trying to predict the center word based on the surrounding words. Instead of creating pairs of input and output tokens, we will create a multi-hot vector of all surrounding terms as an input vector.



How does the network learn the vector representations?

For the sentence

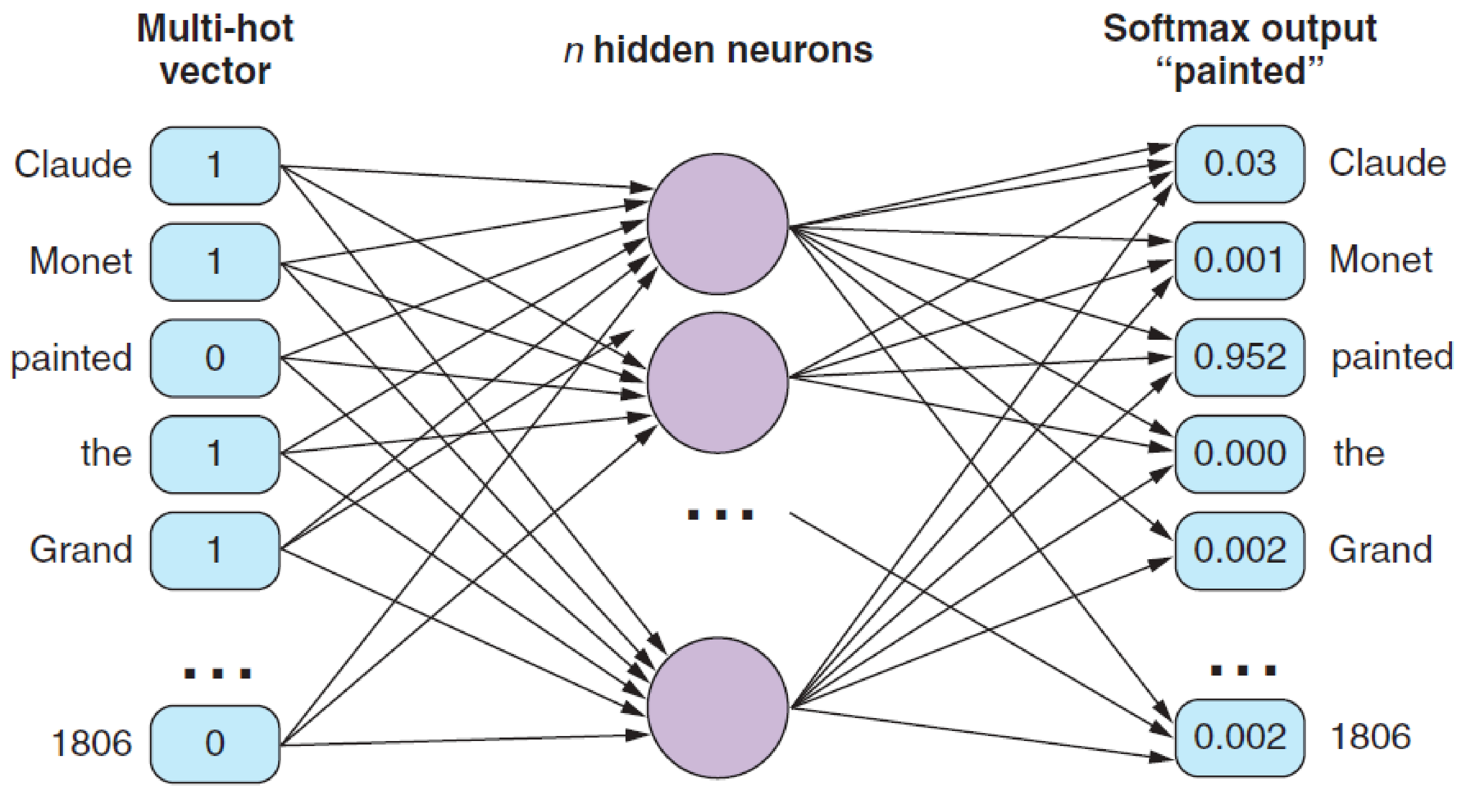
```
sentence = "Claude Monet painted the Grand Canal of Venice in 1806."
```

the 5-grams look like

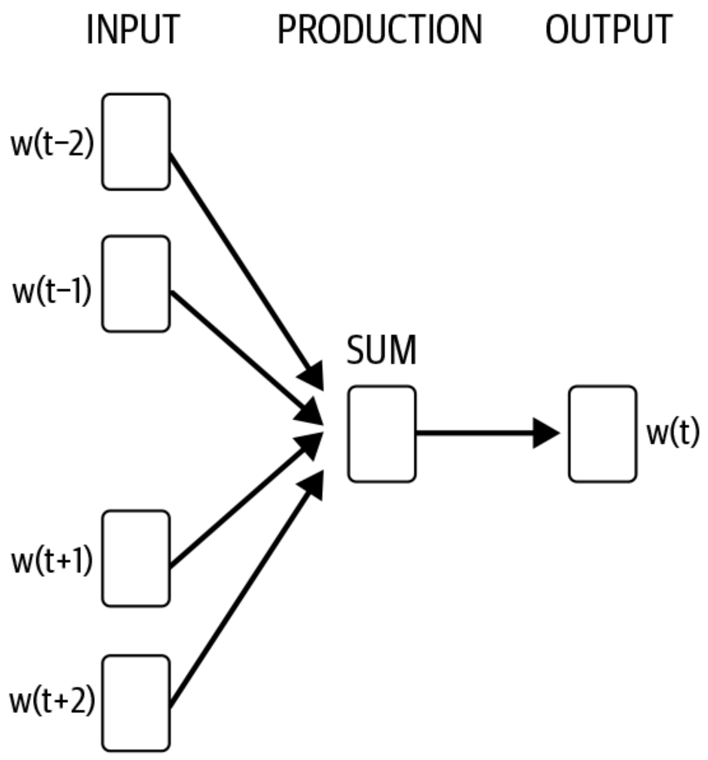
Input word w_{t-2}	Input word w_{t-1}	Input word w_{t+1}	Input word w_{t+2}	Expected output w_t
		Monet	painted	Claude
	Claude	painted	the	Monet
Claude	Monet	the	Grand	painted
Monet	painted	Grand	Canal	the
painted	the	Canal	of	Grand
the	Grand	of	Venice	Canal
Grand	Canal	Venice	in	of
Canal	of	in	1908	Venice
of	Venice	1908		in
Venice	in			1908

Based on the training sets, we can create multi-hot vectors as inputs and map them to the target word as output. The multi-hot vector is the sum of the one-hot vectors of the surrounding words' training pairs. During the training, the output is derived from the softmax of the output node with the highest probability.

How does the network in CBOW look like?



CBOW Model



CBOW